

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12409802
B2
September 09, 2025
Jessup; Chris P. et al.

Combination inflatable and web-based restraint for a motor vehicle seat

Abstract

A combination inflatable and web-based restraint for a motor vehicle seat may include an occupant restraint collar having an arcuate portion with first and second spaced apart legs each joined thereto or integral therewith, the collar configured to extend about a neck and over opposite shoulders of an occupant of the seat, first and second shoulder webs of the restraint each movably coupled to a respective one of the first and second legs such the collar is movable along the first and second shoulder webs, an inflatable airbag housed, in an uninflated state, within the arcuate portion and the first and second legs of the collar, the airbag and configured to open a seam of, and extend out of, the arcuate portion and the first and second legs upon deployment thereof, and at least one inflator fluidly coupled to the airbag for deploying the airbag.

Inventors: Jessup; Chris P. (Sheridan, IN), Bittner; Douglas W. (Indianapolis, IN), Gale; Steven (Colfax, IN)

Applicant: INDIANA MILLS & MANUFACTURING, INC. (Westfield, IN)

Family ID: 85156162

Assignee: INDIANA MILLS & MANUFACTURING, INC. (Westfield, IN)

Appl. No.: 18/578392

Filed (or PCT Filed): August 02, 2022

PCT No.: PCT/US2022/039130

PCT Pub. No.: WO2023/014688

PCT Pub. Date: February 09, 2023

Prior Publication Data

Document Identifier	Publication Date
---------------------	------------------

Related U.S. Application Data

us-provisional-application US 63229078 20210804

Publication Classification

Int. Cl.: B60R21/18 (20060101); B60R21/2165 (20110101); B60R21/261 (20110101); B60R22/00 (20060101)

U.S. Cl.:

CPC B60R21/18 (20130101); B60R21/2165 (20130101); B60R21/261 (20130101); B60R22/00 (20130101); B60R2021/2612 (20130101)

Field of Classification Search

CPC: B60R (21/18); B60R (21/2165); B60R (21/261); B60R (22/00); B60R (2021/2612)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5330255	12/1993	Stawicki	N/A	N/A
6237945	12/2000	Aboud	280/808	B60R 21/18
7926839	12/2010	Mothaffar	280/733	B60R 22/001
8480127	12/2012	Rathmann-Ramlow et al.	N/A	N/A
8714589	12/2013	Santana-Gallego et al.	N/A	N/A
8888127	12/2013	Santana-Gallego	N/A	N/A
9745065	12/2016	Santana-Gallego	N/A	B64D 11/06
10850699	12/2019	Dry et al.	N/A	N/A
2002/0125701	12/2001	Devonport	280/733	B60R 21/18
2002/0171233	12/2001	Grace	280/733	B60R 21/23138
2008/0246264	12/2007	Gerfast	N/A	N/A
2009/0179407	12/2008	Kalandek	280/733	B60R 21/18
2012/0193899	12/2011	Santana-Gallego	280/736	B60R 21/268
2020/0254953	12/2019	Jessup et al.	N/A	N/A

OTHER PUBLICATIONS

PCT International Search Report and Written Opinion completed by the ISA/US on Oct. 6, 2022 and issued in connection with PCT/US2022/039130. cited by applicant
Extended European Search Report for application No. EP22853790.8, dated May 16, 2025. cited by applicant

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This patent application is a U.S. National Stage entry of PCT Application No. PCT/US2022/039130, filed Aug. 2, 2022, which claims the benefit of, and priority to, U.S. Provisional Patent Application Ser. No. 63/229,078, filed Aug. 4, 2021, the disclosures of which are expressly incorporated herein by reference in their entireties.

FIELD OF THE INVENTION

(1) This disclosure relates generally to restraint systems for occupant seats in motor vehicles, and more specifically to such restraint systems including inflatable and web-based restraints.

BACKGROUND

(2) Motor vehicle seat restraint systems typically include a multi-point restraint harness, made up of one or more restraint webs, coupled thereto. Such motor vehicle seat restraint systems are typically incorporated into a motor vehicle seat assembly to be installed in a motor vehicle, with a portion of the restraint system and/or seat assembly being fixedly coupled to the motor vehicle. Some such motor vehicle seat restraint systems may include or incorporate one or more inflatable restraints.

SUMMARY

(3) This disclosure may comprise one or more of the features recited in the attached claims, and/or one or more of the following features and combinations thereof. In one aspect, a combination inflatable and web-based restraint for an occupant of a motor vehicle seat may comprise an occupant restraint collar having an arcuate portion, and first and second spaced apart legs each joined to or integral with the arcuate portion and extending downwardly from the arcuate portion to first and second respective free ends, the arcuate portion configured to extend about a back of a neck and over opposite shoulders of the occupant with the first and second legs extending downwardly along respective sides of a torso of the occupant, a first shoulder web extending through or over a seat back of the vehicle seat, the first shoulder web movably coupled to the first leg adjacent to a front of the seat back such that the collar is movable along the first shoulder web, a second shoulder web extending through or over the seat back and movably coupled to the second leg adjacent to the front of the seat back such the collar is movable along the second shoulder web, an inflatable airbag housed, in an uninflated state, within arcuate portion and the first and second legs of the occupant restraint collar, the airbag and configured to open a seam of, and extend out of, the arcuate portion and the first and second legs of the occupant restraint collar upon deployment thereof, and at least one inflator fluidly coupled to the airbag for deploying the airbag.

(4) In another aspect, an occupant seat for a motor vehicle may comprise a seat bottom configured to be mounted to a motor vehicle, a seat back extending upwardly away from the seat bottom to define an occupant support area between the seat bottom and a front side of the seat back, a restraint harness configured to restrain an occupant seated in the occupant support area, the restraint harness including first and second shoulder webs each extending from the occupant support area through or over the seat back, an occupant restraint collar having an arcuate portion, and first and second spaced apart legs each joined to or integral with the arcuate portion and extending downwardly from the arcuate portion to first and second respective free ends, the arcuate portion configured to extend about a back of a neck and over opposite shoulders of the occupant in the occupant support area with the first and second legs extending downwardly along respective

sides of a torso of the occupant, the first shoulder web movably coupled to the first leg and the second shoulder web movably coupled to the second leg, an inflatable airbag housed, in an uninflated state, within arcuate portion and the first and second legs of the occupant restraint collar, the airbag and configured to open a seam of, and extend out of, the arcuate portion and the first and second legs of the occupant restraint collar upon deployment thereof, and at least one inflator fluidly coupled to the airbag for deploying the airbag.

(5) In yet another aspect, a combination inflatable and web-based restraint for an occupant of a motor vehicle seat may comprise a U-shaped occupant restraint collar configured to extend about a back of a neck and over opposite shoulders of the occupant, a first shoulder web extending through or over a seat back of the vehicle seat, the first shoulder web movably coupled to the collar adjacent to a front of the seat back such that the collar is movable along the first shoulder web, a second shoulder web extending through or over the seat back and movably coupled to the collar adjacent to the front of the seat back such the collar is movable along the second shoulder web, the collar also movable relative to the seat back, an inflatable airbag housed, in an uninflated state, within the collar, the airbag and configured to open a seam of, and extend out of, the collar upon deployment thereof, at least one filler tube configured to extend through, and move relative to, the seat back of the vehicle seat, the at least one filler tube having at least one end fluidly coupled to the airbag within the collar and at least an opposite end disposed along a back side of the seat back, and at least one inflator having at least one gas outlet fluidly coupled to the at least an opposite end of the at least one filler tube, the at least one inflator configured to inflate and deploy the airbag.

(6) In a further aspect, an occupant seat for a motor vehicle may comprise a seat bottom configured to be mounted to a motor vehicle, a seat back extending upwardly away from the seat bottom to define an occupant support area between the seat bottom and a front side of the seat back, a restraint harness configured to restrain an occupant seated in the occupant support area, the restraint harness including first and second shoulder webs each extending from the occupant support area through or over the seat back, a U-shaped occupant restraint collar configured to extend about a back of a neck and over opposite shoulders of the occupant, the collar movably coupled to the first and second shoulder webs, and the collar movable relative to the seat back, an inflatable airbag housed, in an uninflated state, within the collar, the airbag and configured to open a seam of, and extend out of, the collar upon deployment thereof, at least one filler tube extending through, and movable relative to, the seat back of the vehicle seat, the at least one filler tube having at least one end fluidly coupled to the airbag within the collar and at least an opposite end disposed along a back side of the seat back opposite the occupant support area, and at least one inflator having at least one gas outlet fluidly coupled to the at least an opposite end of the at least one filler tube, the at least one inflator configured to inflate and deploy the airbag.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) This disclosure is illustrated by way of example and not by way of limitation in the accompanying figures. Where considered appropriate, reference labels have been repeated among the figures to indicate corresponding or analogous elements.

(2) FIG. 1 is a front elevational view of an embodiment of a motor vehicle seat and occupant restraint system illustrating an embodiment of the occupant restraint system in the form of a combination inflatable and web-based restraint.

(3) FIG. 2 is a front perspective view of the vehicle seat and occupant restraint system illustrated in FIG. 1.

(4) FIG. 3 is a rear perspective view of the vehicle seat and occupant restraint system illustrated in FIGS. 1 and 2.

- (5) FIG. 4 is a rear elevational view of the vehicle seat and occupant restraint system illustrated in FIGS. 1-3.
- (6) FIG. 5A is a front elevational view of a portion of the combination inflatable and web-based restraint illustrated in FIGS. 1-4.
- (7) FIG. 5B is a cross-sectional view of the combination inflatable and web-based restraint as viewed along section lines 5B-5B of FIG. 5A.
- (8) FIG. 6 is a front elevational view of the vehicle seat and occupant restraint system illustrated in FIGS. 1-5B shown with an occupant seated in the vehicle seat and the occupant restraint system engaged over the occupant to restrain the occupant in the seat.
- (9) FIG. 7 is a front elevational view of the vehicle seat and occupant restraint system similar to FIG. 6 showing the inflatable restraint deployed over and about the occupant of the vehicle seat.
- (10) FIG. 8A is a rear elevational view of another embodiment of a motor vehicle seat and occupant restraint system illustrating another embodiment of the occupant restraint system in the form of another combination inflatable and web-based restraint.
- (11) FIG. 8B is a cross-sectional view of the combination inflatable and web-based restraint shown in FIG. 8A as viewed along section lines 8B-8B of FIG. 8A.
- (12) FIG. 9A is a front elevational view of yet another embodiment of a motor vehicle seat and occupant restraint system illustrating yet another embodiment of the occupant restraint system in the form of yet another combination inflatable and web-based restraint.
- (13) FIG. 9B is a rear elevational view of the embodiment of the motor vehicle seat and occupant restraint system illustrated in FIG. 9A.
- (14) FIG. 10 is a rear elevational view of another embodiment of the occupant restraint system.
- (15) FIG. 11A is a rear elevational view of yet another embodiment of the occupant restraint system.
- (16) FIG. 11B is a rear elevational view of a further embodiment of the occupant restraint system.
- (17) FIG. 12 is a rear elevational view of still another embodiment of the occupant restraint system.
- (18) FIG. 13 is a front perspective view of still a further embodiment of the occupant restraint system.
- (19) FIG. 14A is a front elevational view of yet another embodiment of the occupant restraint system.
- (20) FIG. 14B is a rear elevational view of the embodiment of the occupant restraint system illustrated in FIG. 14A.
- (21) FIG. 15 is a front elevational view a further embodiment of the occupant restraint system.
- (22) FIG. 16 is a cross-sectional view of the occupant restraint system of FIGS. 1-7, as viewed along section lines 16-16 of FIG. 1, illustrating an embodiment incorporating a restraint harness pretensioning system.
- (23) FIG. 17 is a front elevational view of another embodiment of the occupant restraint system implemented in a 3-point restraint configuration.
- (24) FIG. 18A is a front perspective view of still a further embodiment of the occupant restraint system.
- (25) FIG. 18B is a rear elevational view of the occupant restraint system of FIG. 18A.
- (26) FIG. 19 is a front elevational view of another embodiment of the occupant restraint system integrated into a modular seat back.

DETAILED DESCRIPTION OF THE DRAWINGS

(27) While the concepts of the present disclosure are susceptible to various modifications and alternative forms, specific exemplary embodiments thereof have been shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that there is no intent to limit the concepts of the present disclosure to the particular forms disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives consistent with the present disclosure and the appended claims.

(28) References in the specification to “one embodiment”, “an embodiment”, “an example embodiment”, etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases may or may not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to effect such feature, structure or characteristic in connection with other embodiments whether or not explicitly described. Further still, it is contemplated that any single feature, structure or characteristic disclosed herein may be combined with any one or more other disclosed feature, structure or characteristic, whether or not explicitly described, and that no limitations on the types and/or number of such combinations should therefore be inferred.

(29) This disclosure relates to the combination of an occupant seat for a motor vehicle and an occupant restraint system coupled to the occupant seat for restraining an occupant of the seat, and further relates to an occupant restraint system including a combination of an inflatable restraint and a web-based restraint. The motor vehicle in which the occupant seat and occupant restraint system is to be mounted may be or include any land-based and/or water-based (e.g., any marine application) motor vehicle or any aircraft, without limitation, and/or may be or include any occupant transportation vehicle that is, or is configured to be, towed or pushed by a motor vehicle.

(30) Referring now to FIGS. 1-4, an embodiment is shown of a combination **10** of an occupant seat **12** and occupant restraint system **14** for a motor vehicle, wherein the occupant restraint system **14** is provided in the form of a combination inflatable and web-based restraint. The occupant seat **12** illustratively includes a seat base **16** configured to be operatively mounted to a floor and/or one or more frame components of a motor vehicle, an occupant seat **18** coupled to the seat base **16** and configured to support an occupant thereon, and a seat back **20** also coupled to the seat base **16** and configured to support a torso of an occupant seated in the occupant seat **18**. A bottom surface **18A** of the occupant seat **18** is coupled to the seat base **16**, and an opposite top surface **18B** of the seat **18** is configured to receive and support an occupant of the seat **12** such that the top surface **18B** of the seat bottom and the front or forward facing surface of the seat back **20** define an occupant support area therealong. A lower or bottom end **20A** of the seat back **20** is mounted to the seat base **16** (and/or in some embodiments to the occupant seat **18**), and the seat back **20** extends upwardly away from the seat **18** to a top end **20B** of the seat back **20**. A front **20C** surface of the seat back **20** faces forward in the motor vehicle, and the opposite surface **20D** defines the back or back surface of the seat back **20**.

(31) In the illustrated embodiment, the seat back **20** includes an occupant support portion **22** having a bottom or lower end **22A** at, just above or just below the top surface **18B** of the occupant seat, and the occupant support portion **22** extends upwardly away from the bottom end **22A** to an opposite top end **22B**. In one embodiment, the top end **22B** is positioned relative to the seat back **20** to be at or near shoulder height of an average occupant of the seat **12**, although in other embodiments the top end **22B** of the occupant support portion **22** may be higher or lower. In any case, a front surface **22C** of the occupant support portion **22** faces forward in the motor vehicle and may be configured, e.g., contoured, to receive and support the back portion of the torso of an average occupant of the seat **12**. The seat back **20** further includes an occupant restraint guide **24** having a lower or bottom end **24A** coupled to the top end **22B** of the occupant support portion **22**, and a top end coupled to the bottom end **26A** of a headrest **26**, wherein the top of the headrest **26** defines the top **20B** of the seat back **20**. In the embodiment illustrated in FIGS. 1-4, the occupant restraint guide **24** defines three openings **28A**, **28B**, **28C** therethrough from the front **20C** of the seat back **20** through the rear **20D** of the seat back. The opening **28A** is illustratively positioned to reside just above the right shoulder of an average occupant of the seat **12**, the opening **28B** is illustratively positioned to reside just above the left shoulder of the average occupant of the seat **12**, and the opening **28C** is illustratively positioned between the openings **28A**, **28B**. In some alternate

embodiments (see FIGS. 8A-9B for examples), the opening 28C may be omitted. In some alternate embodiments in which the seat back 20 is designed such that shoulder webs of the restraint system 14 extend over (not through) the seat back 20, at least the openings 28A, 28B, and in some cases also the opening 28C, may be omitted.

(32) In the embodiment illustrated in FIGS. 1-4, the occupant restraint system 14 includes a combination of an inflatable and web-based restraint, both at least partially housed in an occupant restraint collar 40 positioned along the front 22C of the seat back 20. In the illustrated embodiment, the occupant restraint collar 40 is in the shape of an inverted “U”, having two downwardly-extending and spaced-apart (and approximately parallel) legs 40A, 40B joined at their upper ends by an arcuate section 40C. As depicted by example in FIG. 6, the collar 40 is illustratively sized and shaped such that the arcuate section 40C extends about a rear portion of the neck 60D and over the shoulders 60B, 60C of an average sized occupant 60 of the seat 12, and the legs 40A, 40B extend downwardly from each respective shoulder 60B, 60C along the front portion of the torso 60A of the occupant 60. In alternate embodiments, the section 40C may have a non-arcuate shape and/or the legs 60B, 60C may be non-parallel relative to one another. In one embodiment, the collar 40 is formed of a conventional polymer or combination of polymers, although in alternate embodiments the collar 40 may be form of other conventional materials or of a combination of one or more polymers and other conventional materials.

(33) In the illustrated embodiment, the web-based portion of the inflatable and web-based restraint 14 illustratively comprises a restraint harness including a first shoulder web 30A coupled at one end to a web retractor 32A secured to the seat base 16 adjacent to the back 20D of the seat back 20, and having an opposite end extending from the retractor 32A upwardly along the back 20D of the seat back 20 and through the opening 28A defined through the occupant restraint guide 24. As further illustrated in FIG. 5A, the opposite end of the shoulder web 30A extends from the opening 28A downwardly into an opening 44A in the arcuate portion 40C of the occupant restraint collar 40 and through the leg 40A of the collar 40, and then extends out of the leg 40A through an opening 44B in the free end of the leg 40A and into engagement with tongue or buckle structure 37A. In alternate embodiments, the shoulder web 30A may extend through one or more slots, channels or other web retainers defined or positioned on or along the back, side or front of the leg 40A of the collar 40. The web-based portion of the inflatable and web-based restraint 14, i.e., the restraint harness, further includes a second shoulder web 30B coupled at one end to another web retractor 32B secured to the seat base 16 adjacent to the back 20D of the seat back 20 and spaced apart from the web retractor 32A, and having an opposite end extending from the retractor 32B upwardly along the back 20D of the seat back 20 and through the opening 28B defined through the occupant restraint guide 24. As further illustrated in FIG. 5A, the opposite end of the shoulder web 30B extends from the opening 28B downwardly into an opening 46A in the arcuate portion 40C of the occupant restraint collar 40 and through the leg 40B of the collar 40, and then extends out of the leg 40B through an opening 46B in the free end of the leg 40B and into engagement with buckle or tongue structure 37B. In alternate embodiments, the shoulder web 30B may extend through one or more slots, channels or other web retainers defined or positioned on or along the back, side or front of the leg 40B of the collar 40. In any case, one of the structures 37A, 37B is coupled to a buckle 38 defining a tongue slot therein, and the other is outfitted with a correspondingly configured tongue such that the two structures 37A, 37B can be releasably engaged with one another via the buckle 38. In some alternate embodiments, the web retractor 32A and/or the web retractor 32B may instead be secured to the floor and/or to a frame component of the motor vehicle in which the seat 12 is installed. In some alternate embodiments, the shoulder webs 30A, 30B may be coupled to, or merged into, one end of a single web having an opposite end coupled to a web retractor secured to the seat base 16 or to the floor or frame component of the motor vehicle.

(34) As best shown in FIG. 5A (and also in FIG. 7), the occupant restraint collar 40 illustratively includes a lower or rear compartment 42A which extends about the periphery of the collar 40 and

which faces the front **20C** of the seat back **20** (or faces an occupant when the occupant is seated in the seat **12**), and an upper or front flap or wall **42B** which also extends about the periphery of the collar **40** over the lower or rear compartment **42A** and which faces forward of the vehicle seat **12**. The compartment **42A** and the flap or wall **42B** are illustratively joined together at a seam **42C** which extends about the periphery of the collar **40** adjacent to an outer edge of the collar **40** as illustrated by example in FIG. 5A. In one embodiment, the openings **44A** and **46A** are illustratively formed through the lower or rear compartment **42A**, although in alternate embodiments the openings **44A** and **46A** may be formed by and between the lower or rear compartment **42A** and the flap or wall **42B**. The openings **44** and **46B** are illustratively formed by and between the lower or rear compartment **42A** and the flap or wall **42B**. In the illustrated embodiment, the shoulder webs **30A**, **30B** are not attached or otherwise coupled to the collar **40** but instead pass through the each respective leg **40A**, **40B** such that the collar **40** is movable along, and with respect to, each shoulder web **30A**, **30B**.

(35) The web-based portion of the inflatable and web-based restraint **14**, i.e., the restraint harness, further includes a first lap web **34A** coupled at one end to another web retractor **36A** secured to one side of the seat base **16** adjacent to the seat bottom **18** or to the floor or frame component of the motor vehicle adjacent to the seat bottom **18**, and having an opposite end extending from the retractor **36A** and into engagement with the tongue or buckle structure **37A**. The restraint harness further includes a second lap web **34B** is coupled at one end to yet another web retractor **36B** secured to an opposite side of the seat base **16** adjacent to the seat bottom **18** or to the floor or frame component of the motor vehicle adjacent to the seat bottom **18**, and having an opposite end extending from the retractor **36B** and into engagement with the buckle or tongue structure **37B**. In the illustrated embodiment, the shoulder web **30A** and the lap web **34A** are coupled together via the tongue or buckle structure **37A**, and the shoulder web **30B** and the lap web **34B** are coupled together via the buckle or tongue structure **37B**, such that the shoulder web **30A** and lap web **34A** are together releasably engageable with the shoulder web **30B** and lap web **34B** via the buckle **38**. In alternate embodiments, one or more of the webs **30A**, **30B**, **34A**, **34B** may be individually and separately releasably engageable with the buckle **38**.

(36) The web-based portion of the inflatable and web-based restraint **14**, i.e., the restraint harness, illustrated by example in FIGS. 1-4 is implemented in the form of a four-point restraint system. In alternate embodiments, the restraint harness of the inflatable and web-based restraint **14** may include more or fewer points of restraint, two examples of which are illustrated in FIGS. 17A and 17B and will be described in detail below.

(37) In the embodiment illustrated in FIGS. 1-4, the inflatable portion of the inflatable and web-based restraint **14** illustratively includes a flexible, inflatable airbag **50**, a portion of which is housed within the occupant restraint collar **40** and another portion of which extends through the opening **28C** defined through the occupant restraint guide **24** and downwardly along the back **20D** of the seat back **20** and into engagement with an inflator **54**. As best shown in FIGS. 3 and 4, the airbag **50** includes a single filler tube **50A** having an open lower end **51** extending downwardly into a guide tube **52** and into operative engagement with an airbag inflator **54**. The inflator **54** is electrically actuatable, and is electrically connected to an external control module **58** via a number, *N*, of electrical wires **56**. In the illustrated embodiment, the inflator **54** includes at least one canister of compressed gas and an electrically controlled actuator which is responsive to an activation signal produced by the control module **58** to cause release the pressurized gas into the filler tube **50A** to rapidly inflate the airbag **50** in a conventional manner. The electrically controlled actuator may illustratively be an electrically actuated solenoid, pyrotechnic actuator or other conventional electrically controlled actuator. In alternate embodiments in which the motor vehicle is equipped with a source of pressurized gas (e.g., air), the inflator **54** may be omitted and the electrical wires **56** depicted in FIG. 4 may instead represent an air hose or tube and the control module **58** depicted in FIG. 4 may instead represent a source of pressurized air such that the air hose or tube **56** is

coupled to and between the filler tube **50A** and the source **58** of pressurized gas.

(38) The guide tube **51** is illustratively attached to the back **20D** of the seat back **20** and thus fixed in position relative to the seat **12**. The filler tube **50A**, inflator **54** operatively coupled thereto and wires **56** electrically connected to the inflator **54** are together movable within and relative to the guide tube **52** so that the occupant restraint collar **40** is thereby movable toward and away from the front surface **20C** of the seat back **20**. As an occupant is seated in the seat **12** and fits the restraint **14** over the shoulders and about the torso and engages the buckle **38**, the filler tube **50A** is movable in and/or out of the opening **28C** defined through the occupant restraint guide **24** to accommodate this movement. Likewise, as the motor vehicle is operated under non-deployment conditions, the filler tube **50A** is movable in and/or out of the opening **28C** so as not to impede movement of the shoulder webs **30A**, **30B** in response to the normal movement of the occupant under such conditions.

(39) Referring now to FIG. 7, the airbag **50** illustratively has the same general shape as the occupant restraint collar **40**, e.g., an inverted U-shape, and includes an arcuate portion **50B** which is fluidly coupled to the filler tube **50A**, and a pair of leg portions **50C**, **50D** which extend downwardly away from the arcuate portion **50B**. The airbag **50** is illustratively sized such that the arcuate portion **50B** extends about the back of the occupant's neck **60D** and over each of the occupant's shoulders **60B**, **60C**, and such that the leg portions **50C**, **50D** extend downwardly toward the seat **18** on either side of the occupant's torso **60A**. In the uninflated state of the airbag **50**, it is housed within the occupant restraint collar **40**, with the arcuate portion **50B** residing in the arcuate portion **40C** of the collar **40** and with each leg portion **50C**, **50D** residing in the respective leg portion **40A**, **40B** of the collar **40**. As illustrated by example in FIG. 5A, the filler tube **50A** extends into the collar **40** via an opening **48** centrally located along the rear arcuate portion **40C** of the collar **40**. In one embodiment, the opening **48** is illustratively formed through the lower or rear compartment **42A**, although in alternate embodiments the opening **48** may be formed by and between the lower or rear compartment **42A** and the flap or wall **42B**. As illustrated by example in FIG. 5B, the leg portion **50D** of the airbag **50** and the shoulder web **30B** together extend, one over the other, through the channel **45** defined between the lower or rear compartment **42A** and the flap or wall **42B** of the collar **40**. The leg **40A** of the collar **40** is similarly configured.

(40) FIG. 6 illustrates the occupant restraint **14** in the undeployed state of the airbag **50**, e.g., during normal, non-deployment operation of the motor vehicle. In this mode, the arcuate portion **40C** of the occupant restraint collar extends about the occupant's neck **60D** and shoulders **60B**, **60C** and the legs **40A**, **40B** of the collar extend downwardly toward the seat **18** on either side of the occupant's torso **60A**. FIG. 7 illustrates the occupant restraint **14** in the deployed state of the airbag **50**, e.g., during impact or impending impact, rollover or the like. As the airbag **50** inflates, the growing volume of gas in the airbag components **50B**, **50C**, **50D** causes the seam **42C** between the lower or rear compartment **42A** and the flap or wall **42B** to open, which pushes the flap or wall **42B** toward the occupant **60** such that the flap or wall **42B** surrounds the neck **60D**, shoulders **60B**, **60C** and torso **60A** of the occupant **60**. In one embodiment, the occupant restraint collar **40** is formed of a synthetic polymer such as Nylon®, although in alternate embodiments the occupant restraint collar **40** may be formed of one or more other conventional materials which may be flexible or semi-rigid.

(41) Referring now to FIGS. 8A and 8B, an alternate embodiment is shown of a combination **100** of an occupant seat **12'** and occupant restraint system **14'** for a motor vehicle, wherein the occupant restraint system **14'** is provided in the form of a combination inflatable and web-based restraint. In this alternate embodiment, the occupant seat **12'** is identical in many respects to the occupant seat **12** illustrated in FIGS. 1-7 and described above, and differs from the occupant seat **12** only in the omission of the opening **28C** defined through the occupant restraint guide **24'** of the seat back **20'**.

(42) In the embodiment illustrated in FIGS. 8A and 8B, the shoulder webs **30A'**, **30B'** are illustratively both provided in the form of web tubes, and the filler tube **50A'** of the airbag **50'** is

illustratively provided in the form of a split or “Y” tube with one portion **50A'.sub.1** extending downwardly into operative engagement with the inflator **54**, with another portion **50A'.sub.2** in fluid communication with the filler tube portion **50A'.sub.1** extending through a channel **55** defined through the shoulder web **30B'** and into engagement with the arcuate portion **50B** of the airbag **50**, and with yet another portion **50A'.sub.3** in fluid communication with the filler tube portion **50A'.sub.1** extending through a similar channel defined through the shoulder web **30A'** and also into engagement with the arcuate portion **50B** of the airbag **50**. In this embodiment, the filler tube portion **50A'.sub.1** (and inflator **54** and wiring **56**) is movable within and along the guide tube **52'** as described above so as not to impede movement of the shoulder webs **30A'**, **30B'** in response to the normal movement of the occupant under non-deployment conditions, but the filler tube portions **50A'.sub.2** and **50A'.sub.3** ride along with and within the shoulder webs **30A'**, **30B'** so that only the shoulder webs **30A'**, **30B'** extend through the occupant restraint guide member **24**.

(43) Referring now to FIGS. **9A** and **9B**, another alternate embodiment is shown of a combination **200** of an occupant seat **12'** and occupant restraint system **14''** for a motor vehicle, wherein the occupant restraint system **14''** is provided in the form of a combination inflatable and web-based restraint. In this alternate embodiment, the occupant seat **12'** is identical to the occupant seat **12'** illustrated in FIG. **8A** and differs from the seat **12** illustrated in FIGS. **1-7** in that the opening **28C** defined through the occupant restraint guide **24'** of the seat back **20'** is omitted.

(44) In the embodiment illustrated in FIGS. **9A** and **9B**, the inflator **54** illustrated in FIGS. **4** and **8A** and described above is omitted from the back **20D** of the seat back **20'**, and small inflators are fluidly coupled directly to each of the leg portions of the airbag **50''** and are housed within the occupant restraint collar **40**. As depicted by example in FIG. **9A**, a partial cutaway of the leg **40A** shows a small inflator **202** housed within the leg **40B** of the collar **40** and fluidly coupled directly to the leg portion **50D'** of the airbag **50''**. The inflator **202** is electrically connected to wiring **56B** (e.g., one or more wires) which is routed to the back **20D** of the seat back **20'** along the shoulder web **30B''**. In the illustrated embodiment, the leg **40A** of the collar **40** likewise contains another inflator **202** fluidly coupled directly to the other leg portion of the airbag **50''**, and this inflator is similarly electrically connected to wiring **56A** (e.g., one or more wires) which is routed to the back **20D** of the seat back **20'** along the shoulder web **30A''**. In some alternate embodiments, one of the inflators **202** may be omitted, and in other alternate embodiments one or more additional inflators **202** may be housed within the collar **40** and fluidly coupled directly to the airbag **50''** and electrically connected to the wiring **56A** and/or **56B**. In any case, the airbag **50''** in the embodiment illustrated in FIGS. **9A** and **9B** does not include the filler tube **50A** and instead includes only the arcuate portion fluidly coupled to the two leg portions as described above with respect to FIG. **7**.

(45) In one embodiment, the wiring **56A**, **56B** is illustratively routed to the rear **20D** of the seat back **20'** by securing the wiring **56A**, **56B** to the webs **30A''**, **30B''** respectively. In the embodiment illustrated in FIGS. **9A** and **9B**, for example, a channel member **31A** is affixed to and between front and back surfaces of the shoulder web **30A** to form the shoulder web **30A''**, and the channel member **31A** defines a channel therethrough alongside the web **30A** through which the wiring **56A** is routed to the rear **20D** of the seat back **20'**. Likewise, another channel member **31B** is affixed to and between front and back surfaces of the shoulder web **30B** to form the shoulder web **30B''**, and the channel member **31B** defines a channel therethrough alongside the web **30B** through which the wiring **56B** is routed to the rear **20D** of the seat back **20'**. The wiring **56A** and **56B** are joined to a wiring cable **56** via Y-connection or other suitable connection such that the inflators **202** can be actuated simultaneously during a deployment event.

(46) Referring now to FIG. **10**, another alternate embodiment is shown of a combination **300** of an occupant seat **12** and occupant restraint system **14''** for a motor vehicle, wherein the occupant restraint system **14''** is provided in the form of a combination inflatable and web-based restraint. In this alternate embodiment, the occupant seat **12** is identical to the occupant seat **12** illustrated in FIGS. **1-7**. The occupant restraint system **14''** illustratively differs from the occupant restraint

system **14** illustrated in FIGS. **1-7** and described above only in the configuration of the filler tube **50A''**. In this alternate embodiment, the filler tube **50A''** illustratively includes two or more telescoping filler tube sections which deploy and retract as the shoulder webs **30A**, **30B** are respectively paid out of and taken up into the web retractors **32A**, **32B**. Such telescoping of the filler tube **50A''** illustratively allows the inflator **54** to be secured, e.g., via straps or brackets **302A**, **302B** to a frame component **304** of the seat back **20**. In the illustrated embodiment, the filler tube **50A''** includes four fluidly-coupled telescoping sections **308**, **310**, **312** and **314** with the section **308** fluidly coupled to the airbag **50** (as described above), the section **314** fluidly coupled to a gas outlet of the inflator **54** and the sections **310** and **310** sequentially fluidly coupled therebetween. It will be understood that in alternate embodiments, the filler tube **50A''** may include more or fewer such telescoping sections. Although not shown in FIG. **10**, it will be further understood that the inflator **54** will be electrically connected to one or more wires for electrically activating the inflator **54** as described above.

(47) Referring now to FIG. **11A**, another alternate embodiment is shown of a combination **400** of an occupant seat **12** and occupant restraint system **14.sup.IV** for a motor vehicle, wherein the occupant restraint system **14.sup.IV** is provided in the form of a combination inflatable and web-based restraint. In this alternate embodiment, the occupant seat **12** is identical to the occupant seat **12** illustrated in FIGS. **1-7**. The occupant restraint system **14** illustratively differs from the occupant restraint system **14'''** illustrated in FIG. **10** and described above only in the configuration of the bottom section of the filler tube **50A'''**. In this alternate embodiment, the filler tube **50A'''** illustratively includes two or more telescoping filler tube sections which deploy and retract as the shoulder webs **30A**, **30B** are respectively paid out of and taken up into the web retractors **32A**, **32B** as described above with respect to FIG. **10**, with the last telescoping section **402** illustratively provided in the form of an inverted "Y" so as to provide for two gas inlets each configured to be operatively coupled to a respective one of two separate inflators **54A**, **54B**. Such telescoping of the filler tube **50A'''** illustratively allows the inflator **54A**, **54B** to be secured, e.g., via straps or brackets **404**, **406** respectively to the frame component **304** of the seat back **20**. In the illustrated embodiment, the filler tube **50A'''** includes four fluidly-coupled telescoping sections **308**, **310**, **312** as described above with the section **310** fluidly coupled to the airbag **50**, and the section **402** having a single outlet **402A** fluidly coupled to the section **312** and two separate inlets **402B**, **402C** configured to be fluidly coupled to a respective one of the inflators **54A**, **54B**. It will be understood that in alternate embodiments, the filler tube **50A'''** may include more or fewer such telescoping sections. Although not shown in FIG. **11A**, it will be further understood that the inflators **54A**, **54B** will both be electrically connected to one or more wires for electrically activating the inflators **54A**, **54B**.

(48) Referring now to FIG. **11B**, yet another alternate embodiment is shown of a combination **500** of an occupant seat **12** and occupant restraint system **14.sup.V** for a motor vehicle, wherein the occupant restraint system **14.sup.V** is provided in the form of a combination inflatable and web-based restraint. In this alternate embodiment, the occupant seat **12** is identical to the occupant seat **12** illustrated in FIGS. **1-7**. The occupant restraint system **14.sup.V** illustratively differs from the occupant restraint system **14.sup.IV** illustrated in FIG. **11A** and described above in that the telescoping filler tube is bifurcated into two separate telescoping filler tube sections **50A1.sup.IV** and **50A2.sup.IV**. The filler tube sections **50A1.sup.IV**, **50A2.sup.IV** are illustratively separate from each other, although in some embodiments the filler tube sections **50A1.sup.IV**, **50A2.sup.IV** may be joined to one another at some point prior to fluidly coupling to the airbag **50**. In any case, each filler tube section **50A1.sup.IV**, **50A2.sup.IV** illustratively extends along the seat back **20D** into fluid communication at one end with a respective one of two separate inflators **54A**, **54B**, and each extends through the opening **28C** defined through the web guide **24** and into fluid communication with the airbag **50**, such that the airbag **50** is inflated via two separate inflators **54A**, **54B** each through a separate, telescoping filler tube section **50A1.sup.IV**, **50A2.sup.IV**. In this alternate

embodiment, the filler tube sections **50A1.sup.IV**, **50A2.sup.IV** each illustratively include two or more telescoping filler tube sections which deploy and retract as the shoulder webs **30A**, **30B** are respectively paid out of and taken up into the web retractors **32A**, **32B** as described above with respect to FIG. **10**. Such telescoping of the filler tube sections **50A1.sup.IV** and **50A2.sup.IV** illustratively allows the inflators **54A**, **54B** to be secured, e.g., via straps or brackets **404**, **406** respectively to the frame component **304** of the seat back **20**. In the illustrated embodiment, each filler tube sections **50A1.sup.IV**, **50A2.sup.IV** includes four fluidly-coupled telescoping sections **308A** and **308B**, **310A** and **310B**, **312A** and **312B** and **314A**, **314B** respectively, with the sections **308A** and **308B** each fluidly coupled to the airbag **50**, and with the sections **314A** and **314B** each fluidly coupled to a fluid outlet of a respective one of the inflators **54A**, **54B**. It will be understood that in alternate embodiments, the filler tube section **50A1.sup.IV** and/or the filler tube section **50A2.sup.IV** may include more or fewer such telescoping sections. Although not shown in FIG. **11B**, it will be further understood that the inflators **54A**, **54B** will both be electrically connected to one or more wires for electrically activating the inflators **54A**, **54B**.

(49) Referring now to FIG. **12**, still another alternate embodiment is shown of a combination **600** of an occupant seat **12** and occupant restraint system **14.sup.VI** for a motor vehicle, wherein the occupant restraint system **14.sup.VI** is provided in the form of a combination inflatable and web-based restraint. In this alternate embodiment, the occupant seat **12** is identical to the occupant seat **12** illustrated in FIGS. **1-7**. The occupant restraint system **14.sup.VI** illustratively differs from the occupant restraint system **14.sup.V** illustrated in FIG. **11B** and described above in that the bifurcated filler tube sections **50A1.sup.IV** and **50A2.sup.IV** are replaced with separate filler tubes **50A1.sup.V**, **50A2.sup.V** each riding on a respective shoulder web **30A**, **30B**, and in that the web guide **24'** is modified to omit the central passageway **28C**. The filler tube **50A1.sup.V** illustratively rides on top of the shoulder web **30A** and extends, along with the shoulder web **30A**, through the opening **28A** defined through the web guide **24'** and into fluid communication at one with the airbag **50**, and the filler tube **50A2.sup.V** illustratively rides on top of the shoulder web **30B** and extends, along with the shoulder web **30B**, through the opening **28B** defined through the web guide **24'** and into fluid communication at one end with the airbag **50**. Each filler tube **50A1.sup.V**, **50A2.sup.V** also extends along the respective shoulder web **30A**, **30B** along the seat back **20D** and into fluid communication at a respective opposite end with a respective one of two separate inflators **54A**, **54B**, such that the airbag **50** is inflated via two separate inflators **54A**, **54B** each through a separate one of the filler tubes **50A1.sup.V**, **50A2.sup.V**. In this alternate embodiment, the filler tubes **50A1.sup.V**, **50A2.sup.V** each illustratively include two or more telescoping filler tube sections which deploy and retract as the shoulder webs **30A**, **30B** are respectively paid out of and taken up into the web retractors **32A**, **32B** as described above with respect to FIG. **10**. Such telescoping of the filler tubes **50A1.sup.V** and **50A2.sup.V** illustratively allows the inflators **54A**, **54B** to be secured, e.g., via straps or brackets **404**, **406** respectively to the frame component **304** of the seat back **20**. In the illustrated embodiment, each filler tube **50A1.sup.V**, **50A2.sup.V** includes four fluidly-coupled telescoping sections **308A** and **308B**, **310A** and **310B**, **312A** and **312B** and **314A**, **314B** respectively, with the sections **308A** and **308B** each fluidly coupled to the airbag **50**, and with the sections **314A** and **314B** each fluidly coupled to a fluid outlet of a respective one of the inflators **54A**, **54B**. It will be understood that in alternate embodiments, the filler tube **50A1.sup.V** and/or the filler tube **50A2.sup.V** may include more or fewer such telescoping sections. Although not shown in FIG. **11B**, it will be further understood that the inflators **54A**, **54B** will both be electrically connected to one or more wires for electrically activating the inflators **54A**, **54B**. In alternate embodiments, one or both of the filler tubes **50A1.sup.V**, **50A2.sup.V** may be non-telescoping.

(50) Referring now to FIG. **13**, another alternate embodiment is shown of a combination **700** of an occupant seat **12** and occupant restraint system **14.sup.VII** for a motor vehicle, wherein the occupant restraint system **14.sup.VII** is provided in the form of a combination inflatable and web-

based restraint. In this alternate embodiment, the occupant seat **12** is identical to the occupant seat **12** illustrated in FIGS. **1-7**. The occupant restraint system **14**.sup.VII illustratively differs from the occupant restraint system **14** illustrated in FIGS. **1-7** and described above in that the filler tube **50A** is replaced with filler tubes **50A1**.sup.VI and **50A2**.sup.VI each riding on and along a respective lap web **34A**, **34B** along a front of the seat **12**, the inflator(s) is/are located on an underside of the seat bottom **18** or secured to the seat base **16**, and the web guide **24'** is modified to omit the central passageway **28C**. The filler tube **50A1**.sup.VI illustratively rides on top of or along the lap web **34A** and extends along the shoulder web **30A** and into fluid communication at one with the portion of the airbag **50** housed within a lower portion of the respective leg of the collar **40**, and the filler tube **50A2**.sup.VI illustratively rides on top of or along the lap web **34B** and extends along the shoulder web **30B** and into fluid communication at one end with the airbag **50** housed within a lower portion of the respective leg of the collar **40**. Opposite ends of the filler tubes **50A1**.sup.VI, **50A2**.sup.VI extend into fluid communication with a single inflator **54**, or into fluid communication with a respective one of two separate inflators **54A**, **54B**, coupled to the seat base **16** and/or seat bottom **18**. In the illustrated embodiment, the filler tubes **50A1**.sup.VI, **50A2**.sup.VI are illustratively each formed of single tubes, although in alternate embodiments the filler tube **50A1**.sup.VI and/or the filler tube **50A2**.sup.VI may be telescoping as described above. In some alternate embodiments, the occupant restraint system **14**.sup.VII may include only one of the filler tubes **50A1**.sup.VI or **50A2**.sup.VI fluidly coupled to and between the airbag **50** and the one or more inflators.

(51) Referring now to FIGS. **14A** and **14B**, another alternate embodiment is shown of a combination **800** of an occupant seat **12** and occupant restraint system **14**.sup.VIII for a motor vehicle, wherein the occupant restraint system **14**.sup.VIII is provided in the form of a combination inflatable and web-based restraint similar to the occupant restraint system **14'** illustrated in FIGS. **9A** and **9B** and described above. In this alternate embodiment, the occupant seat **12'** is identical to the occupant seat **12'** illustrated in FIGS. **9A-9B**, and the occupant restraint system **14**.sup.VIII includes only a single inflator **202** coupled to one leg portion **50D'** of the airbag **50'''** within the one respective leg of the collar **40**. The opposite leg of the collar **40**, in this alternate embodiment, does not include an inflator coupled to the airbag **50'''**. The inflator **202** is electrically connected to wiring **56B** (e.g., one or more wires) routed to the back of the seat back along the shoulder web **30B''** as described above with respect to FIGS. **9A** and **9B**.

(52) Referring now to FIG. **15**, another alternate embodiment is shown of a combination **900** of an occupant seat **12** and occupant restraint system **14** for a motor vehicle, wherein the occupant restraint system **14** may be implemented in any of the forms illustrated in the attached figures and described herein which include the occupant restraint collar **40**. The embodiment illustrated in FIG. **15** differs from any such occupant restraint systems only in that buttons **802A**, **802B** are illustratively attached to, e.g., sewn onto, the respective shoulder webs **30A**, **30B** below each respective leg of the collar **40**. The buttons **802A**, **802B** illustratively act as stops to the respective legs of the collar **40** to prevent the collar **40** from being displaced too far downwardly relative to the seat **12**. It will be understood that the buttons **802A**, **802B**, and their locations relative to the respective lower ends of the collar **40**, are provided only by way of example, and that other conventional structures may alternatively be coupled to or between one or both of the shoulder webs **30A**, **30B** and the collar **40** to prevent displacement of the collar **40** below a selected position relative to the seat **12** bottom or relative to the web coupling buckle.

(53) Referring now to FIG. **16**, yet another alternate embodiment is shown of a combination **1000** of an occupant seat **12** and occupant restraint system **14** for a motor vehicle, wherein the occupant seat **12** and the occupant restraint system **14** may be implemented in any of the forms illustrated in the attached figures and described herein which include the occupant restraint collar **40** or any of its variants. The embodiment illustrated in FIG. **16** differs from other embodiments of the combination occupant seat **12** and occupant restraint system **14** illustrated in the attached figures and described

herein by the inclusion of an occupant restraint pretensioning module **1002** mounted to the seat base **16** and/or seat bottom **18**, and operatively coupled to the occupant restraint system **14**. Operation of the pretensioning module **1002** is conventional in that it is configured to apply a specified or variable amount of tension to and on the restraint system **14** to tighten the lap webs **34A**, **34B** and shoulder harnesses **30A**, **30B** about the occupant of the seat **12**. In embodiments in which the seat **12** is a suspension seat, the pretensioning module **1002** may further be configured to apply a specified or variable amount of tension to and on the restraint system **14** to, following pretensioning of the restraint system **14**, pull the suspension seat downwardly toward, and in some embodiments into contact with the seat base **16** and/or floor of the motor vehicle.

(54) In the illustrated embodiment, the pretensioning module **1002** includes at least one engagement member **1004**, e.g., in the form of at least one hook, clip, bracket or the like, configured to be secure, via at least one web engagement member **35A**, **35B**, e.g., in the form of at least one hook, clip, bracket or the like, to at least one of the webs **30A**, **30B**, **34A**, **34B** and/or to at least one tongue or buckle of the restraint system **14**. In one example embodiment, the engagement member(s) **1004** and the web engagement member(s) **35A**, **35B** may each include a single web engagement member configured to be secured to one another and with the web engagement member **35A** operatively coupled to, i.e., secured to, one of the lap webs **34A**, **34B** or to one of the shoulder webs **30A**, **30B** or to one tongue or buckle of the restraint system **14**. In other embodiments, the engagement member(s) **1004** and the web engagement member(s) **35A**, **35B** may each include two web engagement members each configured to be secured to one another and with the web engagement members **35A**, **35B** each operatively coupled to, i.e., secured to, a respective one of the lap webs **34A**, **34B**, to a respective one of the shoulder webs **30A**, **30B** or to one of the shoulder webs **30A**, **30B** and one of the lap webs **34A**, **34B** or to two tongues or buckles of the restraint system **14**.

(55) In any case, the occupant restraint pretensioning module **1002** is illustratively configured to be electrically connected to at least one decision-making circuit, e.g., to the external control module **58** illustrated by example in FIG. 4, and programmed to selectively activate the pretensioning module **1002** to apply tension to one or more of the webs **30A**, **30B**, **34A**, **34B**, e.g., in response to (e.g., prior to or as part of) a vehicle impact event, e.g., a frontal, side or rear impact, a vehicle rollover event or other vehicle event, or in response to an occupant request, e.g., via occupant activation of a manual switch electrically coupled to the decision-making circuit.

(56) Referring now to FIG. 17, yet another alternate embodiment is shown of a combination **1100** of an occupant seat **12''** and occupant restraint system **14.sup.IX** for a motor vehicle, wherein the occupant seat **12''** differs from the occupant seats **12**, **12'** in that it does not include the web guide **24** and wherein the occupant restraint system **14.sup.IX** differs from the other embodiments illustrated in the attached figures and described therein in that the restraint harness (comprising a 4-point restraint) is implemented in the form of a conventional 3-point restraint harness with a single inflatable airbag housed within a sleeve **40'** operatively coupled to the shoulder web portion. In the illustrated embodiment, the restraint harness includes a single web **30'** having one end extending over and about the upper portion of the seat back and coupled to a rear portion of the seat back, and having an opposite end attached to the seat bottom or seat base. The web **30'** passes through a web guide **1102** which partitions the web **30'** into a shoulder web **30** and a lap web **34**, wherein the web guide **1102** is secured to a releasable engagement member **1104** configured to releasably engage another engagement member **1106** attached to the seat bottom and/or seat base. In the illustrated embodiment, the engagement member **1104** is a conventional tongue member and the engagement member **1106** is a conventional buckle member, with each configured to releasably engage one another in a conventional manner, although in alternate embodiments the engagement member **1104** may be a buckle member and the engagement member **1106** may be a tongue member. In the illustrated embodiment, the fill tube **50A.sub.P** runs along the back side of the shoulder portion **30** of the web **30'**, and fluidly couples at one end to the airbag **50** carried within the sleeve **40'** and

fluidly couples at an opposite end to an inflator secured to the rear of the seat back as described above. In some embodiments, the fill tube **50A.sub.P** may be a single length of tubing, although in alternate embodiments the fill tube **50A.sub.P** may include two or more telescoping segments as described above. In other alternate embodiments, the portion of the shoulder web **30** extending between the top of the sleeve **40'** and the inflator may be provided in the form of a web tube as illustrated by example in FIGS. **8A** and **8B** and described above, wherein the filler tube **50A.sub.P** extends through a channel defined through the shoulder web **30** and into engagement with the inflator and with the airbag carried by and within the sleeve **40'**.

(57) Referring now to FIGS. **18A** and **18B**, still another alternate embodiment is shown of a combination **1200** of an occupant seat **12"** and occupant restraint system **14.sup.X** for a motor vehicle, wherein the occupant restraint system **14.sup.X** is provided in the form of a combination inflatable and web-based restraint. In this alternate embodiment, the occupant seat **12"** and the occupant restraint system **14.sup.X** are both similar to the respective occupant seat **12** and occupant restraint system **14** illustrated in FIGS. **1-7** but each differs in that the collar **40"** of the occupant restraint system **14.sup.X** is secured to, or integral with, the seat back **20** of the occupant seat **12"**. In the illustrated embodiment, the U-shaped portion of the collar **40"** is secured to or integral with the web guide **24** such that the collar **40"** is not movable relative to the seat **12"** or relative to the shoulder webs **30A**, **30B** of the restraint system **14.sup.X** which pass through the collar **40"** as described above with respect to the collar **40**. The webs **30A**, **30B**, **34A**, **34B** are otherwise as described hereinabove. The collar **40"**, and the airbag **50** carried thereby, are otherwise as described above with respect to the collar **40** and airbag **50**.

(58) Because the collar **40"** is secured to the seat back **20** and is therefore not movable with or along the shoulder webs **30A**, **30B**, the filler tube **50.sup.VII** extending through the opening **28C** of the web guide **24** into fluid communication with the airbag **50** is likewise not movable relative to the seat back **20** and is instead secured to the seat back **20**, seat bottom **18** or seat base **16** via a tether **1202** or other mechanical attachment mechanism. In the illustrated embodiment, for example, the tether **1202** is coupled to one end of the inflator **54** having an opposite end fluidly coupled to the filler tube **54A.sup.VII**. In alternate embodiments, the filler tube **50.sup.VII** may be secured directly to the seat back **20**, seat bottom **18** or seat base **16** via a tether, strap, bracket or other conventional securement device(s).

(59) Referring now to FIG. **19**, a further alternate embodiment is shown of a combination **1300** of an occupant seat **12.sup.IV** and occupant restraint system **14** for a motor vehicle. Illustratively, the seat **12.sup.IV** is identical to the seat **12** illustrated in FIGS. **1-7** except that the seat bottom **18** and seat base are omitted, and the web retractors **32A**, **32B**, **36A** and **36B** are all mounted to the back surface **20D** of the seat back **20**. The restraint system **14** is otherwise identical in structure and operation to the restraint system **14** illustrated in FIGS. **1-7** and described above, although in alternate embodiments the restraint system **14** may be provided in the form of any of the variants of the restraint system **14** illustrated in the attached figures and described above.

(60) The seat **12.sup.IV** and restraint system **14** together illustratively define a mobile or modular combination **1300** of an occupant seat **12.sup.IV** and occupant restraint system **14** which may be secured to a wall or other structure of a motor vehicle above, and usable with, an existing occupant seat or bench **1302**. Although intended for used with patient-transporting emergency vehicles, e.g., ambulances and the like, it will be understood that the combination **1300** may be used in any motor vehicle which includes an occupant seating structure and a wall and/or other structure extending upwardly from the combination **1300** to which the combination **1300** can be mounted.

(61) While the disclosure has been illustrated and described in detail in the drawings and foregoing description, such an illustration and description is to be considered as exemplary and not restrictive in character, it being understood that only illustrative embodiments have been shown and described and that all changes and modifications consistent with the disclosure and recited claims are desired to be protected.

Claims

1. A combination inflatable and web-based restraint for an occupant of a motor vehicle seat, the combination comprising: an occupant restraint collar having an arcuate portion, and first and second spaced apart legs each joined to or integral with the arcuate portion and extending downwardly from the arcuate portion to first and second respective free ends, the occupant restraint collar configured to engage the occupant with the arcuate portion extending about a back of a neck and over opposite shoulders of the occupant and with the first and second legs extending downwardly along respective sides of a torso of the occupant, a first shoulder web configured to extend through or over a seat back of the vehicle seat, the first shoulder web extending through the first leg adjacent to a front of the seat back, a second shoulder web configured to extend through or over the seat back, the second shoulder web extending through the second leg adjacent to the front of the seat back, an inflatable airbag housed, in an uninflated state, within the arcuate portion and the first and second legs of the occupant restraint collar, the airbag configured to open a seam of, and extend out of, the arcuate portion and the first and second legs of the occupant restraint collar upon deployment thereof, and at least one inflator fluidly coupled to the airbag for deploying the airbag, wherein, with the entire occupant restraint collar engaging the occupant, the entire occupant restraint collar is movable relative to the motor vehicle seat along each of the first and second shoulder webs.
2. The combination of claim 1, wherein the combination further comprises at least one filler tube configured to extend through, and move relative to, the seat back of the vehicle seat, the at least one filler tube having at least one end fluidly coupled to the airbag within the collar and at least one opposite end fluidly coupled to at least one gas outlet of the at least one inflator disposed along a back side of the seat back.
3. The combination of claim 2, further comprising an elongated guide tube configured to be mounted to the back side of the seat back with an opening facing a top of the seat back, the elongated guide tube configured to receive the at least one inflator and at least a portion of the filler tube therein via the opening such that the at least one inflator and the at least a portion of the guide tube are movable within and relative to the guide tube with movement of the at least one filler tube relative to the seat back.
4. The combination of claim 2, wherein the at least one inflator comprises a first inflator and a second inflator each having a gas outlet fluidly coupled to the at least one opposite end of the at least one filler tube.
5. The combination of claim 2, wherein the at least one filler tube comprises at least two filler tube segments configured for telescoping movement relative to one another.
6. The combination of claim 2, wherein at least a portion of the first shoulder web between the collar and the at least one inflator comprises a first web tube and at least a portion of the second shoulder web between the collar and the at least one inflator comprises a second web tube, and wherein the at least one filler tube comprises: a first filler tube disposed within the first web tube and configured to extend with the first web tube through or over the seat back, and a second filler tube disposed within the second web tube and configured to extend with the second web tube through or over the seat back, wherein the first and second filler tubes are each fluidly coupled to and between the airbag and the at least one inflator.
7. The combination of claim 2 wherein the at least one filler tube comprises: a first filler tube configured to ride on and extend with the first web through or over the seat back, and a second filler tube configured to ride on and extend with the second web through or over the seat back, wherein the first and second filler tubes are each fluidly coupled to and between the airbag and the at least one inflator.
8. The combination of claim 6, wherein the at least one inflator comprises a first inflator having a

gas outlet fluidly coupled to the first filler tube and a second inflator having a gas outlet fluidly coupled to the second filler tube.

9. The combination of claim 6, wherein the first filler tube comprises at least first and second filler tube segments configured for telescoping movement relative to one another, and wherein the second filler tube comprises at least third and fourth filler tube segments configured for telescoping movement relative to one another.

10. The combination of claim 1, wherein the at least one inflator is configured to be mounted to a seat bottom or a seat base of the motor vehicle seat, and wherein the combination further comprises at least one filler tube configured to extend along at least one lap web and to be fluidly coupled to and between the airbag and at least one gas outlet of the at least one inflator.

11. The combination of claim 10, wherein the at least one filler tube comprises at least two filler tube segments configured for telescoping movement relative to one another.

12. The combination of claim 1, wherein the at least one inflator comprises at least one inflator disposed within the collar and fluidly coupled to the airbag within the collar, and wherein the combination further comprises: wiring operatively coupled to the at least one inflator and extending along at least one of the first and second shoulder webs and extending with the at least one of the first and second shoulder webs over or through the seat back, and at least one channel member affixed to and between front and back surfaces of, and extending along, the at least one of the first and second shoulder webs to define at least one channel therebetween, wherein the wiring extends through the at least one channel.

13. The combination of claim 1, further comprising means for preventing displacement of the collar below a selected position relative to a seat bottom of the motor vehicle seat.

14. An occupant seat for a motor vehicle, the occupant seat comprising a seat bottom configured to be mounted to the motor vehicle, a seat back extending upwardly away from the seat bottom to define an occupant support area between the seat bottom and a front side of the seat back, and the combination inflatable and web-based restraint of claim 1, wherein the first and second shoulder webs are part of a restraint harness configured to restrain an occupant seated in the occupant support area, the first and second shoulder webs each extending from the occupant support area through or over the seat back.

15. The occupant seat of claim 14, wherein the at least one inflator is disposed along a back side of the seat back opposite the occupant support area, and further comprising at least one filler tube extending through, and movable relative to, the seat back of the vehicle seat, the at least one filler tube having at least one end fluidly coupled to the airbag within the collar and at least an opposite end fluidly coupled to at least one gas outlet of the at least one inflator.

16. The occupant seat of claim 15, further comprising an elongated guide tube mounted to the back side of the seat back with an opening facing a top of the seat back, the elongated guide tube receiving the at least one inflator and at least a portion of the filler tube therein via the opening such that the at least one inflator and the at least a portion of the guide tube are movable within and relative to the guide tube with movement of the at least one filler tube relative to the seat back.

17. A combination inflatable and web-based restraint for an occupant of a motor vehicle seat, the combination comprising: a U-shaped occupant restraint collar configured to extend about a back of a neck, and over opposite shoulders, and downwardly along opposite sides of a torso of, the occupant, a first shoulder web configured to extend through or over a seat back of the vehicle seat, the first shoulder web extending through the collar adjacent to a front of the seat back, a second shoulder web configured to extend through or over the seat back, the second shoulder web extending through the collar adjacent to the front of the seat back such the collar is movable relative to the motor vehicle seat along each of the first and second shoulder webs, an inflatable airbag housed, in an uninflated state, within the collar, the airbag configured to open a seam of, and extend out of, the collar about a periphery of the collar upon deployment thereof, the inflatable air bag including at least one filler tube configured to extend through the seat back of the vehicle seat,

the at least one filler tube having at least one end fluidly coupled to the airbag within the collar such that the at least one filler tube moves with the collar relative to the motor vehicle seat, and at least an opposite end configured to be disposed along a back side of the seat back, and at least one inflator having at least one gas outlet fluidly coupled to the at least an opposite end of the at least one filler tube, the at least one inflator configured to inflate and deploy the airbag.

18. The combination of claim 17, further comprising an elongated guide tube configured to be mounted to the back side of the seat back with an opening facing a top of the seat back, the elongated guide tube configured to receive the at least one inflator and at least a portion of the filler tube therein via the opening such that the at least one inflator and the at least a portion of the guide tube are movable within and relative to the guide tube with movement of the at least one filler tube relative to the seat back.

19. An occupant seat for a motor vehicle, the occupant seat comprising: a seat bottom configured to be mounted to the motor vehicle, a seat back extending upwardly away from the seat bottom to define an occupant support area between the seat bottom and a front side of the seat back, a restraint harness configured to restrain an occupant seated in the occupant support area, the restraint harness including first and second shoulder webs each extending from the occupant support area through or over the seat back, a U-shaped occupant restraint collar configured to extend about a back of a neck, over opposite shoulders, and downwardly along opposite sides of a torso of, the occupant, the collar movably coupled to the first and second shoulder webs, and the collar movable relative to the occupant seat along the first and second shoulder webs, an inflatable airbag housed, in an uninflated state, within the collar, the airbag configured to open a seam of, and extend out of, the collar about a periphery of the collar upon deployment thereof, the inflatable airbag including at least one filler tube extending through the seat back of the vehicle seat, the at least one filler tube having at least one end fluidly coupled to the airbag within the collar such that the at least one filler tube moves with the collar along the first and second webs and relative to the motor vehicle seat, and at least an opposite end disposed along a back side of the seat back opposite the occupant support area, and at least one inflator having at least one gas outlet fluidly coupled to the at least an opposite end of the at least one filler tube, the at least one inflator configured to inflate and deploy the airbag.

20. The occupant seat of claim 19, further comprising an elongated guide tube mounted to the back side of the seat back with an opening facing a top of the seat back, the elongated guide tube receiving therein the at least one inflator and at least a portion of the filler tube via the opening such that the at least one inflator and the at least a portion of the guide tube are movable within and relative to the guide tube with movement of the at least one filler tube relative to the seat back.
